

Exam practice

October 13, 2025

INTELLIGENT AGENTS

- a) Define Artificial Intelligence, Machine Learning, and Deep Learning, highlighting their key characteristics and differences.
- b) Clarify what "percepts" and "commands" are and how they fit into the architecture of an intelligent agent. Explain the distinction between the belief state and percept history.

SEARCH

- a) Between depth first search and breadth first search, which will find the shortest path through a maze, and why? What are the main differences between the two algorithms?
- b) What are the characteristics of an admissible and consistent heuristic?

LOGIC

- a) Explain what syntax and semantics are in the context of propositional logic. How are propositions constructed? What values can they take on?
- b) Why is first-order logic considered to be more expressive than propositional logic? What are the main differences between the two frameworks?

CONSTRAINTS SAT PROBLEMS AND OPTIMIZATION

- a) For what type of problems and search space would you use a local search approach? What are the limitations of a steepest ascent/descent algorithm?
- b) Specify how node consistency and arc consistency relate to CSPs and provide an explanation of how they can be enforced to solve the constraint satisfaction problem.

UNCERTAINTY

- a) What is conditional probability? How does it relate to the joint probability distribution?
- b) What is the Markov Assumption? What are the two probability types present in Hidden Markov Models? What do they represent?

MACHINE LEARNING

- a) Machine learning includes different tasks, such as supervised, unsupervised, and reinforcement learning. How do these tasks differ?
- b) What is overfitting? Can you name a strategy to avoid it?

ETHICS AND SOCIAL IMPACT

- a) Discuss common biases in Artificial Intelligence and provide examples.